An Empirical Study on Developing Stacking Ensemble Model for Bangla Sports Sentiment Analysis

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***Abstract*—Bangladeshi people have a strong connection with sports, evident in the multitude of comments made before and after games regarding their favorite teams. Sports sentiment analysis is conducted to understand public sentiments in the realm of sports. In this work, we proposed a new sports based sentiment dataset extracted from YouTube comments covering football, cricket, traditional local games and others, totaling 82,612 instances which is publicly available in the repository https://github.com/cseku170202/Bangla-Sports-Sentiment-Analysis. Our empirical study considered the use of various trendy machine learning (ML) and deep learning (DL) methods. Specifically, we utilized Convolutional Neural Network (CNN) and Transformer Neural Network (TNN) as deep learning methods and Multinomial Naive Bayes (MNB), Random Forest (RF), and K-Nearest Neighbors (KNN) as machine learning techniques. Furthermore, we examined 48 different stacking ensemble models by combining these machine learning and deep learning algorithms. Subsequently, our proposed stacking model achieved the highest accuracy of 81%, demonstrating its capability to identify sports sentiments with remarkable F1-score of 0.82. To explain model's interpretability, we utilized several Explainable AI techniques to better understand the impact of the base model in the stacking model output.**

***Index Terms*—Bangla Dataset, Sports Sentiment Analysis, Stacking Ensemble Model, Explainable AI, SHAP.**

I. Introduction

Sentiment analysis is the practice of examining people's beliefs, feelings, and attitudes about many topics, including events, businesses, societal issues, and products [1]. Sentiment analysis from Bangla text data is very popular now because of the increasing use of social media to express people's thoughts [2], [3]. The population of Bangladesh demonstrates a tendency for expressing their sentiments predominantly via social media platforms, manifesting in various forms such as comments within game interfaces, video commentary sections, and social media posts [4], [5]. Approximately 250 million individuals converse in Bangla as their primary language, with 160 million of these speakers residing in Bangladesh [6]. A significant majority of individuals conversant in Bangla employ the Bangla language and its associated characters to articulate their thoughts across various social media platforms. Within the realm of sports, one can discern the sentiments of Bangla individuals towards their favored teams or players, evident through diverse comments that reflect their unique perspectives [7]. Sentiment analysis has emerged as a focal point in the domain of Natural Language Processing (NLP), directing attention toward techniques that extract pertinent insights from textual data, capturing the nuanced expressions of emotions, feelings, and opinions conveyed by individuals [5]. The process of sentiment analysis is grouping textual material according to its underlying sentiment polarity—positive, negative, or neutral [6], [8]. While individual sports like cricket and football have been investigated independently, there is a notable scarcity of research addressing sentiment analysis in the broader context of Bangla sports [9]–[13]. Our sentiment analysis uniquely hones in on the diverse emotional expressions found in Bangla-language YouTube video comments, specifically delving into the realm of sports, with a particular focus on the popularity of football, cricket, and other sports, including those that may be rooted in rural or indigenous traditions within Bangladesh. The prospect of conducting sentiment analysis on authentic human-generated Bangla text data is a source of excitement and intrigue for our research endeavors.

The NSC boasts affiliation with a comprehensive array of 46 distinct sports federations. Cricket and Football reign as the most popular sports, with Ha-Du-Du being recognized as the national game [14]. Sports sentiment analysis finds application in various domains [15]. It enables us to analyze the performance or public sentiment surrounding a particular match, utilizing sentiment analysis allows us to predict the potential winner of any match [16]. Sports sentiment analysis has a wide range of applications outside of the sports world, including in society, security, travel, banking, business, healthcare, and entertainment, among other sectors .

The majority of research has been on multi-class sentiment prediction, classifying attitudes into positive, negative, and neutral classifications using Machine Learning (ML) or Deep Learning (DL) methods. A machine learning algorithm is a computational approach that utilizes input data to perform a designated task, eliminating the need for explicit, hard-coded instructions to generate a predetermined outcome [17]. Deep learning (DL) represents a burgeoning field in machine learning (ML) research, characterized by the incorporation of multiple hidden layers within artificial neural networks . This work focuses on determining the fundamental sentiments (i. e., positive and negative) of game reviews from Bangla social media data. The following is a summary of the main contributions made by this work:

1) Created a new text dataset from sports-related YouTube videos, encompassing 82,612 feedback entries. The sentiment analysis is anchored in a novel dataset encompassing a spectrum of popular games in Bangladesh, spanning cricket, football, and even local games.
2) Developed 48 different ensemble methods (Stacking Models) integrating Machine Learning and Deep Learning Model.
3) Interpret the best model using various methods from explainable AI.

The paper proceeds as follows: Related works are presented in section II. Our proposed methodology is described in section III. Section V wraps up our study, while section IV provides an analysis and discussion of the experimental results.

II. Related Work

Although there are many studies on Bangla sentiment analysis, very little analysis has been done on Bangla text collected from sports. Below are some related works that have been done on sports sentiment analysis for Bangla and other languages.

In the year of 2022, M. Rabeya et al. [9] introduced a model by gathering 4061 data points from existing sports-related Facebook pages and groups. Analyzing the data using various deep learning models, including CNN, LSTM, and BiLSTM, revealed that the CNN model achieved the highest accuracy of 94.57%. A year before this in 2021, M. A. Faruque et al. [10] gathered reviews of Bangla cricket from different verified Facebook pages for Bangla sentiment analysis and took another dataset (ABSA) [18] and applied different ML algorithms. In this LR performs very well and provides 83% accuracy. Same year in 2021, K. Tirdad et al. [11] presented an approach collecting data from Twitter. Where they assembled a total of 82,842 tweets and provided an automated deep neural network approach to analyze the sentiments of common people. They employed 12 distinct systems, each leveraging artificial neural networks with varying parameters. The top seven models were combined to create an ensemble system. The authors recorded highest 62.71% f1-score and 62.72% accuracy for their proposed system. In the same year and another resource in 2021, using machine learning methods including SVM, naïve Bayes (NB) and LR, a categorization method developed by M. Alqmase [12] which can automatically categorize Arabic text against sports enthusiasm. With 3688 tweets in the test data set and 86,794 anti-fanatic and fanatic tweets in the training data set, LR produced the best accuracy of 91%. In 2021, A. Barua et al. [13] worked on sports news text. She gathered 43,306 news documents and six of the most widely used machine learning techniques (e.g., Decision Tree (DT), Random Forest (RF), Support Vector Classifier (SVC), Multinomial Naive Bayes (MNB) etc.). Of these, Support Vector Classifier (SVC) yielded the highest accuracy, of 97.60%.

As we can see in previous studies [9], [11], [13], their dataset consisted of 4061, 2979, 43306 comments/reviews respectively for Bangla language. Here we can note that there is a crisis of Bangla sports related rich dataset. To the best of our knowledge we have created the largest Bangla sports related sentiment dataset consisting of 82,612 instances. In our proposed approach, we aim to build upon these discoveries to address a gap in understanding within this particular research field. The relevant works are summarized in Table I.

III. Proposed Methodology

This study's main goal is to develop a model that can identify whether Bangla comments on sports videos convey positive or negative sentiments. Figure 1 shows the sequential steps for the development of the proposed model.

A. Dataset Preparation

In this research, Bangla comments were gathered from sports videos on YouTube utilizing web scraping methods. The sourced Bangla comment data encompasses contents from cricket, football, other global sports, and local sports YouTube videos. The comment data was amassed within an Excel file and subsequently categorized into positive and negative sentiments through manual annotation. The

Table I
SUMMARY OF THE PREVIOUS WORKS ON SPORTS SENTIMENT ANALYSIS

Name	Year	Dataset Description	Proposed Model	Accuracy	F1-Score	Advantages	Disadvantages
M. Rabeya [9]	2022	4061 (Facebook comment)	CNN	94.57%	-	The functioning of each layer within the model is demonstrated.	Dataset indicates low volume
M. A. Faruque [10]	2021	2501 (New Data) + 2979 (ABSA dataset [18])	LR	83.23% (New Data) & 77.68% (ABSA Data)	-	Two different datasets were used.	The two datasets used were too small.
K. Tirdad [11]	2021	82,842 (Tweets Data)	Ensemble system (FFNN+ single-layer CNN+Multi-layer CNN+GRU+LSTM +Bi-Dir-LSTM+TCN)	-	62.72%	Seven different deep learning-based Artificial neural networks were used for twelve distinct systems.	F1 score is not very good
M. Alqmase [12]	2021	177276 (Tweets Data)	LR	-	91.1%	Created a large and balance data.	Utilized only two classification categories.
A. Barua [13]	2021	43306 (Document)	SVC + unigram + bigram + trigram	-	97.60%	ML models were employed for classification tasks, utilizing TF-IDF values of different n-grams	The corpus they created exhibits imbalance.

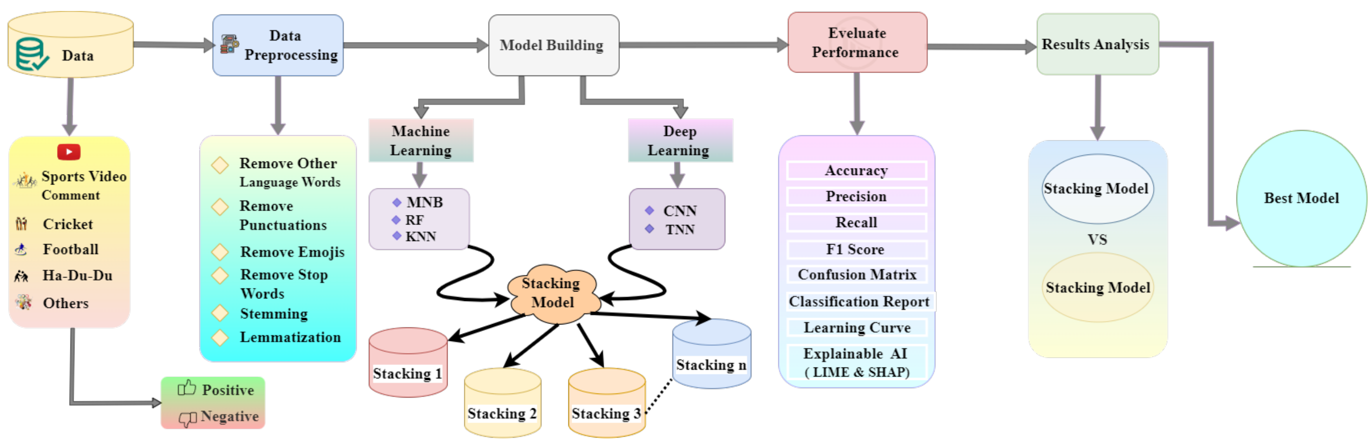

Figure 1. Sequential Steps for the Development of the Proposed Model

proposed dataset of Bangla sports video comments comprises a total of 82,612 entries, encompassing 41,256 positive comments and 41,356 negative comments, indicating a relatively balanced distribution within the dataset. We have given public access to the proposed novel dataset [1].

B. Data Preprocessing

Data preprocessing involves the removal of irrelevant characteristics or features from the dataset in order to enhance its quality and suitability for analysis in research [15]. The collected data may contain many types of noise, such as non-Bangla words, emojis, punctuations, stop words, and many more, which reduce the performance of the model. Various data pre-processing techniques are used, like non-Bangla words removal, emoji removal, removing punctuations, stop words removal, stemming and lemmatization, etc.

[1]https://github.com/cseku170202/Bangla-Sports-Sentiment-Analysis

1) Non-Bangla words removal, emoji removal, removing punctuations: : We have removed the punctuation mark or special symbol from our dataset. The punctuation marks and emojis that have been removed are (”’““£|¢|Ñ+-*/=EROer012–34567 • 89 !()-[];:’”“, ˙#$%&̂_˜”). We also removed non-Bangla words that were present in our dataset.

2) Stop words removal: Stop words are words that appear often in sentences but don't really add anything to the context or general meaning of the statement [3]. Such as “ও”(and), “এই” (this), “টি” (the) etc. We extracted 363 stop words from our proposed dataset and eliminated them to get a reduced fresh version of the dataset. The comment “এই মার্টিনালি ভালো খেলে” (this Martinali plays well) and the comment after removing stop words is “মার্টিনালি ভালো খেলে”.

3) Stemming: The root word of a word is taken through stemming [3]. For example, “বাংলা” (Bangla), “বাংলাদেশে” (Bangladesh), “বাংলাদেশি”(Bangladeshi) the root word of these words is “বাংলা” (Bangla). The Bangla comment: “তিনি দীর্ঘদিনের পর সম্প্রতি সাইক্লিং শুরু করেছেন” (he has recently started cycling after a long time) after stemming the Bangla comment: "তিনি দীর্ঘদিন পর সম্প্রতি সাইক্লিং শুরু করেছেন".

4) Lemmatization : Lemmatization is an advanced form of stemming which transforms each word into its basic form conducting both morphological and non-morphological operations [3]. For example, Bangla comment "ক্রিকেটাররা মাঠে অনুশীলন করছে" (cricketers are practicing on the field) become "ক্রিকেটার মাঠ অনুশীলন কর" after performing the lemmatization.

C. Feature Extraction

Learning models always require a numerical representation of the raw data where a unique numeric representation is given for each unique word of the dataset [8]. This process is known as feature extraction or vectorization. We utilized CountVectorizer to extract features for the machine learning model. Using CountVectorizer, a set of text documents may be transformed into a matrix of token counts, with each row representing a document and each column representing a distinct word within the dataset [8]. In the process of feature extraction for deep learning algorithms such as CNN and TNN, we also employed another feature extraction method named word embedding. We employed a tokenizer to break down textual data into individual words. These words are then converted into sequences of integers through tokenization. Following tokenization, the integer sequences undergo padding to ensure a consistent length using the pad sequence method.

D. Sports Sentiment Analysis Model

We employed various machine learning and deep learning algorithms for sports sentiment analysis, including Multinomial Naive Bayes (MNB), Random forest classifier(RF), and K-Nearest Neighbors (KNN) as machine learning algorithms, and Convolutional Neural Network (CNN) and Transformer Neural Network (TNN) as deep learning algorithms. The layers that we mainly used in CNN algorithm are the embedding layer, Convolutional layer, max pooling layer and dense layer. For the TNN architecture, our model comprised of layers including the input layer, embedding layer, SimpleTransformerBlock layer, GlobalMaxPooling1D layer, and dense layer. Ultimately, different stacking model was created by utilizing various combinations of machine learning and deep learning models. The detailed results analysis of the deployed models will be thoroughly discussed in section IV of this article.

IV. Experimental Results Analysis & Discussion

The main objective of our proposed work is to create a model that can accurately analyze the sports sentiments of Bangla speaking community. For this we have used various machine learning and deep learning algorithms, afterwards we developed various stacking models based on different combination of machine learning and deep learning algorithms for more accurate sports sentiment analysis.

A. Model Selection (ML and DL)

Three popular machine learning methods for sentiment analysis in Bangla were assessed in this study: Random Forest (RF), Multinomial Naive Bayes (MNB), and K-Nearest Neighbors (KNN). Convolutional Neural Network (CNN) and Transformer Neural Network (TNN) models are two deep learning methods that we employed for sports sentiment analysis. The accuracy overview of the implemented base algorithms is given in Table II whereas Table III presents a comparison of deep learning and machine learning models in terms of accuracy. The CNN model has the best accuracy (79%), while the MNB model has the lowest accuracy (64%), as shown by the comparison in Table II.

Table II
Accuracy overview of the implemented base algorithms

SN	Model	Accuracy
1	MNB	0.64
2	RF	0.72
3	KNN	0.66
4	CNN	**0.79**
5	TNN	0.78

B. Development of Stacking Ensemble Models

We have developed 48 different stacking models using various combinations of machine learning and deep learning models to improve the interpretation of sports sentiment where the final estimators were MNB, RF, KNN, and LR. A comparison of the accuracy of stacking models is shown in Table III. It is evident that Logistic Regression (LR) produces superior results than other final estimators when comparing them for the same base model. The maximum accuracy is provided by several stacking models (MNB+CNN+TNN), (RF+CNN+TNN), (TNN+CNN), (RF+KNN+TNN), (MNB+RF+TNN) and (MNB+RF+KNN+CNN+TNN) with an accuracy of 0.81 whereas LR is the final estimator.

Table III
Accuracy Comparison of Different Stacking Models

SN	Stacking Model	Accuracy			
		Final Estimators			
		LR	RF	MNB	KNN
1	MNB+CNN+TNN	**0.81**	0.80	0.48	0.77
2	RF+CNN+TNN	**0.81**	0.80	0.66	0.79
3	KNN+CNN+TNN	0.80	0.78	0.63	0.78
4	MNB+RF+KNN	0.76	0.75	0.49	0.74
5	TNN+CNN	**0.81**	0.78	0.64	0.77
6	MNB+RF+KNN+CNN+TNN	**0.81**	0.80	0.58	0.78
7	RF+KNN+CNN	0.78	0.77	0.61	0.77
8	RF+KNN+TNN	**0.81**	0.78	0.56	0.78
9	MNB+KNN+CNN	0.79	0.77	0.59	0.76
10	MNB+KNN+TNN	0.78	0.74	0.52	0.75
11	MNB+RF+CNN	0.80	0.78	0.48	0.77
12	MNB+RF+TNN	0.81	0.79	0.48	0.78

For multiple stacking models, around same accuracy (0.81) was attained. In order to choose the best model

Table IV
COMPARISON AMONG BEST PERFORMED STACKING MODELS

SN	Stacking Model	Accuracy	F1-Score
1	MNB+CNN+TNN	0.81	0.81
2	RF+CNN+TNN	0.81	**0.82**
3	TNN+CNN	0.81	0.81
4	MNB+RF+KNN+CNN+TNN	0.81	0.81
5	RF+KNN+TNN	0.81	0.81
6	MNB+RF+TNN	0.81	0.81

out of the group, Table IV compares the F1-score of the top 6 models. It is clear that the (RF+CNN+TNN) stacking model has the greatest F1 score, of 0.82. Thus, the stacking model we suggest is (RF+CNN+TNN).

The classification report of the (RF+CNN+TNN) stacking model is shown in Table V for a better understanding of the model's performance. Furthermore the binary classifier system's diagnostic capability is shown by the ROC curves depicted in Figures 2 and 3 respectively, which changes with the discrimination threshold. With an AUC of 0.90 for both positive and negative classes, the model exhibits excellent discrimination, as shown in Figure 2. This indicates high true positive rates and low false positive rates, respectively, for the negative class. With a macro-average AUC of 0.90, the stacking model performs well overall and guarantees accurate classification in all classes. In contrast, Figure 3 illustrates the differing discriminative capacities of the fundamental models, Random Forest (RF), Convolutional Neural Network (CNN), and Transformer Neural Network (TNN). AUCs for the CNN and RF models are strong at 0.87, however TNN's AUC is a little bit lower at 0.86. Overall, the discrimination shown by the CNN and RF models is equivalent and strong, with the former slightly outperforming the latter.

Table V
CLASSIFICATION REPORT FOR (RF+CNN+TNN) STACKING MODEL

	Precision	Recall	F1 Score	Support
Negative	0.83	0.80	0.81	8345
Positive	0.80	0.83	0.82	8109
Accuracy			0.81	16454
Macro-Avg	0.81	0.81	0.81	16454
Weighted-Avg	0.81	0.81	0.81	16454

Table VI
VARIOUS METRICS FOR (RF+CNN+TNN) STACKING MODEL

SN	Metrics	Evaluation
1	MSE	0.18
2	MAE	0.18
3	RMSE	0.42
4	AUC	0.90

Table VI provides a better picture of the performance of the proposed stacking model in sports sentiment analysis by displaying the different error matrices of the model, including MSE, MAE, RMSE, and AUC.

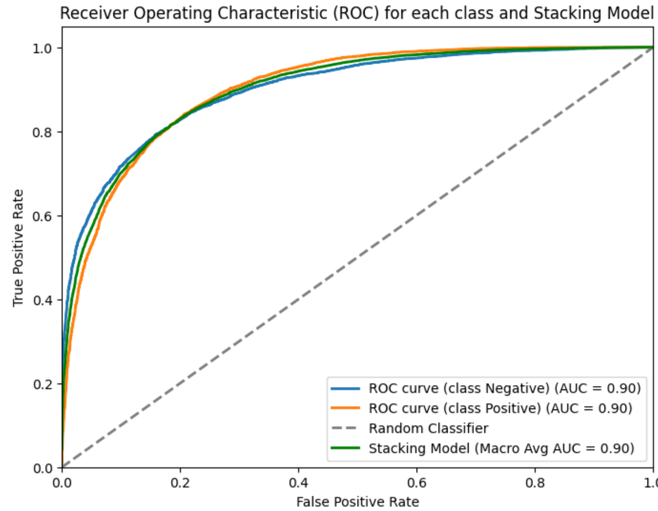

Figure 2. ROC Curve for the (RF+CNN+TNN) Stacking Model with Class Label

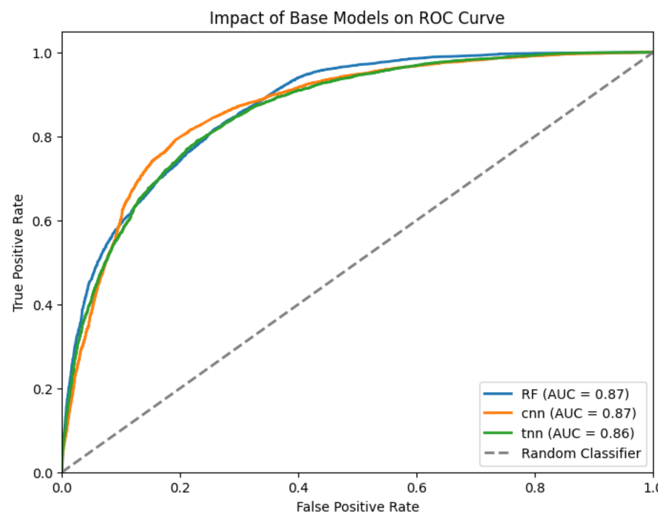

Figure 3. ROC Curve for the (RF+CNN+TNN) Stacking Model with Base Models

We can observe from the error matrix, which includes MSE, MAE, and RMSE, that the error of our suggested stacking model is 0.25, 0.49, and 0.49, respectively while the stacking model also showed the greatest AUC value of 0.90, at the same time. From the confusion matrix of Figure 4 we can get a clear idea about how well our proposed stacking model classifies positive and negative sentiments. The proposed stacking model accurately classified 6843 positive sentiments and misclassified 1266. On the other hand, the proposed model accurately classified 6573 negative sentiments and misclassified 1772 instances.

C. Explainable AI

Explanatory artificial intelligence (AI) refers to a set of approaches and a taxonomy system that are designed to

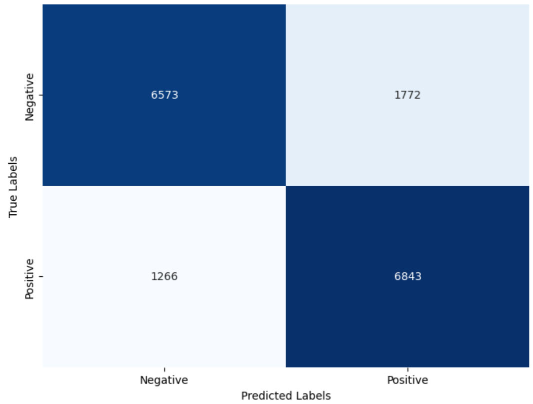

Figure 4. Confusion Matrix for ((RF+CNN+TNN) Stacking Model

analyze and clarify the complex workings of models [19], [20]. From *Explainable AI* to interpret the stacking model, we have retrieved Local Interpretable Model-agnostic Explanation (LIME) and SHapley Additive exPlanations (SHAP) modules in this study. LIME can explain which category is more likely to be selected using its prediction probabilities and shows how each attribute contributes to document classification [19]. The LIME explanation for an example sentence is shown in Figure 5. Sample text "জাতীয় সংগীত জোস হই" (the national anthem has been fantastic) is a positive sentiment and from LIME explanation we can observe that the probability of a positive forecast is 0.78 and that of a negative prediction is 0.22. To understand

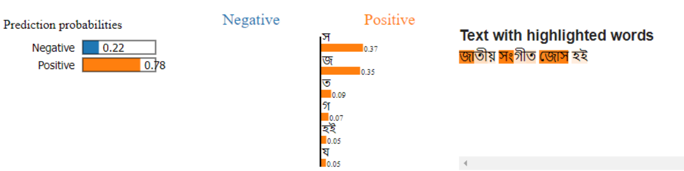

Figure 5. LIME Explanation for the (RF+CNN+TNN) Stacking Model

the impact of the feature value on the output of the stacking model, we extracted the stacking SHAP value and plotted it through a summary plot (Figure 6) where the x axis is along the SHAP value and along the y axis is the feature value with feature index. From the SHAP summary plot we can observe that the feature value of feature index 184 has the most impact on the output of the model and the feature index 166 has the least impact. Where feature index 184 is "অমল" and feature index 168 is "অবল ".

V. Conclusion

Our main aim in this study is to analyze the sentiments of Bangla people regarding sports. The excitement of Bangla speaking community in any game is always noticeable. Now, it is reflected in social media. For sports sentiment analysis, we have developed a dataset

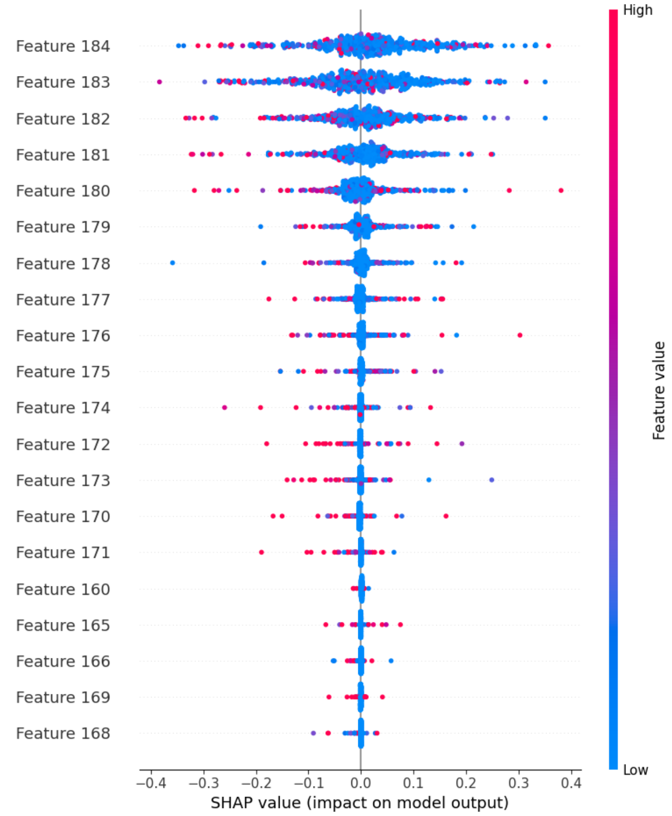

Figure 6. SHAP Feature Importance for the (RF+CNN+TNN) Stacking Model

containing 82,612 entries from sports-related YouTube videos and provided public access of the dataset. Afterwards, we apply various machine learning and deep learning algorithms on the proposed dataset. Where machine learning algorithms include Multinomial Naive Bayes (MNB), Random forest classifier (RF), and K-Nearest Neighbors (KNN), and deep learning includes Convolutional Neural Networks (CNN) and Transformer Neural Networks (TNN). Furthermore, we examined 48 different stacking models with different combinations of this machine learning and deep learning model. We identified (RF+CNN+TNN) as the best stacking model by comparing its F1-score with other stacking models that have the highest accuracy of 0.81. The F1-score of our suggested stacking model (RF+CNN+TNN) was 0.82. So, we are optimistic that our proposed stacking model, RF+CNN+TNN, will perform very well in sports sentiment analysis. In the future, we will try to enrich and balance our dataset more, implement various advanced feature extraction techniques.

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